

AI-Assisted Contract Review for a Small In-House Legal Team

Solution design — Jianhua (Catherine) Chen, 7 July 2026

1. Reframing the problem

The brief describes a throughput problem: a small legal team, a steady stream of low-stakes commercial contracts (mutual NDAs, small order forms), a queue that outgrows the day, and painful spikes at month- and quarter-end.

But the underlying problem is a **mismatch between effort and risk**: every contract gets the same full human read, even though ~80% are boilerplate and most redlines are ones the team has answered before. The queue is a symptom; uniform review of non-uniform risk is the cause.

That reframing matters because it tells us what AI should do here. The goal is not "AI reads contracts instead of lawyers." It is:

Route every contract to the cheapest review path that is safe for its risk level — and let AI do the first pass so lawyers review the AI's work instead of reading everything from scratch.

Two second-order effects are worth naming:

- **Contracts block revenue.** At quarter-end the legal queue sits directly in the sales close path, so cutting turnaround time has commercial value well beyond legal-team hours.
- **Repetition is an asset.** "Edits we've seen before" means the team already has an implicit playbook. Codifying it is the highest-leverage step in this whole project.

2. What I'd validate before building

I would spend the first days answering these with the legal team (assumptions below are labelled as assumptions, not facts):

1. **Volume and mix** — contracts per week by type; how spiky is the end-of-quarter peak?
2. **Where the time actually goes** — reading clean contracts, negotiating redlines, or chasing context from sales? Each points to a different bottleneck.
3. **What "safe to fast-track" means to them** — which contract types, value thresholds and clause deviations they'd never want auto-cleared.
4. **The paper trail** — do we have historical redline exchanges (Word tracked changes, email threads)? This is the raw material for the playbook.
5. **Current tooling** — is there a CLM in place, or is intake ad hoc via email? This decides whether we build a thin layer or integrate into an existing system.

Assumptions are placeholders, not commitments. The specifics used in this document and in the demo — a shared legal inbox as the intake channel, "template v3.2", the contract mix, the SLA and sampling rates — are illustrative, chosen to keep the design concrete. Week 1 validates each against how the business actually operates, and the design holds if they change: the pipeline needs *an* intake channel, not that particular one.

3. Proposed solution: a risk-based triage pipeline

STEP 1 · ARRIVAL — ONE FRONT DOOR

Every incoming contract enters through a single channel — the shared legal inbox, the contract system, or a feed from the sales tools (whichever matches how contracts actually arrive; see the assumptions note above). Each contract is timestamped on arrival and its waiting time stays visible in the queue, so nothing can quietly pile up.



STEP 2 · FIRST LOOK — WHAT IS THIS?

The AI reads the document and answers four questions: what kind of contract is this? Who is the other party? Is it on our own standard template, or on their paperwork? And how confident is it in those answers? If it is not confident, or the type is unfamiliar, the contract goes straight to full human review (Tier 3) — it never guesses.



STEP 3 · COMPARISON — WHAT CHANGED?

The system compares the contract against our standard template, clause by clause. The matching itself is an exact text comparison — no AI judgement involved. The AI then answers one question per difference found: does it change anyone's rights or obligations, or is it cosmetic (names, dates, formatting)? A second, independently-run AI pass repeats every judgement; if the two disagree about any clause, the contract moves down a tier — disagreement is a warning light, never ignored.



TIER 1 · CLEAN COPY

Matches our template with no meaningful changes. The lawyer gets a short verification report and signs off in seconds. Later — once accuracy is proven with real data — these can clear automatically, always with random spot-checks.

TIER 2 · FAMILIAR EDITS

The other side has made edits we have seen before. For each edit the AI looks up the team's playbook and proposes: accept, counter with pre-approved wording, or hand to the lawyer — plus a drafted reply email. The lawyer reviews a prepared package, not a blank page.

TIER 3 · NEW OR UNCERTAIN

Anything unusual, unfamiliar or low-confidence lands here. The AI writes a short orientation note — what this is, what to look at — but makes no recommendations. A lawyer reads the contract in full. This is the safety valve.



STEP 4 · HUMAN SIGN-OFF — ALWAYS

A lawyer personally reviews and approves everything before it leaves the company — the system never sends anything to a counterparty on its own. Every step is recorded (what the AI checked, what the lawyer changed, which playbook version applied), and a random slice of cleared contracts is silently re-queued for a blind second review.

The playbook loop — how the system learns, under control: when the same new situation keeps appearing (repeated escalations, lawyers rewriting the AI's suggestions), the system drafts a playbook addition and queues it as a proposal. Only the playbook owner can approve a change, so the company negotiates with one voice — and the playbook version increments, never rewriting history.

(The demo in [2-demo/index.html](#) walks through one contract in each tier, plus a live version of this map and an "Audit & sampling" view.)

Tier 1 — clean contracts (the volume killer). The contract matches our own template with no material deviations. AI performs a clause-by-clause comparison and issues a short verification report ("matches standard template; 2 immaterial differences: date

format, party address"). A lawyer signs off in seconds instead of minutes. Once trust is earned (§5), this tier can auto-clear.

Tier 2 — familiar redlines (the judgement multiplier). The counterparty sent edits. AI maps each edit to the playbook: *accept* (within our fallback position), *counter* (drafts our standard counter-language), or *escalate* (no playbook position exists). It drafts the response email and the counter-redlined document. The lawyer reviews a prepared package rather than starting from a blank page.

Tier 3 — everything else (the safety valve). Novel clause types, unusual counterparties, high contract value, or low model confidence → full human review, with an AI-generated issue summary as a head start. **Anything the system is unsure about lands here by default.** The failure mode we design against is a risky contract being mis-tiered *down*, so tiering thresholds are conservative and the legal team sets them.

The playbook is the real product. For each contract type: our standard positions, acceptable fallbacks per clause, hard red lines, and approved counter-language. Without it, LLM output is generic and unverifiable — and lawyers rightly won't trust it. With it, every AI suggestion is checkable against a position the team itself wrote. To build it, I'd mine 50–100 historical negotiations to draft it, then have the lawyers correct and own it. The AI's job is to *apply* the playbook consistently, not to have legal opinions of its own. ([3-worked-example/](#) shows this grounding in practice; [4-prompts/](#) shows how the playbook is injected into every prompt.)

Playbook governance: one company position, with room to learn

- **Two permission levels.** *Reviewers* (the lawyers) use the playbook, propose changes, and keep a personal layer. The *playbook owner* (e.g. the senior lawyer / GC) approves global changes and sets the tiering thresholds. Small team, so this is a maker-checker rule, not a bureaucracy.
- **One global playbook holds all negotiation positions.** Deliberately, lawyers do **not** get personal playbooks for positions: if each reviewer counters the same clause differently, the company's negotiation stance becomes inconsistent, counterparties get different terms depending on who reviews, and the audit story collapses. Consistency is the point of having a playbook at all.
- **What is personal:** drafting style and email tone preferences, plus a personal *staging area* — a lawyer can trial a draft position on their own reviews (always flagged in the UI as "trial position — not company standard") and then propose it for promotion to global.

- **Rewrites are contract-local.** When a reviewer replaces a proposed counter with their own wording, the change applies to that contract only — the playbook text and all historical records are untouched, and versions only ever move forward. Recurring rewrites of the same kind surface automatically as a proposal for the owner to consider.
- **Iteration loop.** Proposals flow in from three streams: recurring escalations of the same clause type, lawyer edits to AI-drafted counters, and sign-off overrides. The owner reviews the proposal queue weekly (ten minutes), approves or rejects, and every change is versioned — so any past AI suggestion stays traceable to the exact playbook version that produced it.

4. Where AI is used — and where it is deliberately not

STEP	AI ROLE	HUMAN ROLE
Intake classification	Classify type, extract metadata	Spot-check; owns taxonomy
Template comparison	Clause-level diff + deviation report	Sign-off on Tier 1
Redline analysis	Map edits to playbook, draft counters + response email	Review, edit, send — accountable for every outgoing document
Escalation triage	Flag novel/low-confidence items	Full review of Tier 3
Playbook maintenance	Propose new entries when the same "novel" edit recurs	Approve every playbook change

Deliberate non-goals for v1: no AI-initiated negotiation, no auto-sending of anything external, no touching contract types outside the pilot scope, and no removal of legal sign-off until the data supports it.

The review surface: what the lawyer actually sees

Triage only earns trust if the reviewing screen makes verification effortless. Three rules (all visible in the demo):

1. **The full contract is always on screen.** Never just the AI's excerpts — summaries sit *above* the document, not instead of it. The overview strip states which template (and version) it matched, how many edits and at what severity, whether anything is novel,

and the key commercial facts (parties, term, dates, governing law) — with those facts highlighted in the source text itself.

2. **Annotations sit beside the text they refer to.** Every edit is highlighted in the contract body; its assessment card sits alongside at the same position — why it needs attention, the playbook position it maps to, how we've resolved similar edits before (clickable precedents), and the proposed replacement text. The lawyer accepts the suggestion or rewrites it in place; their rewrite feeds the playbook proposal queue.
3. **Output is composed for dispatch, not copy-paste.** Once the edits are resolved, the response email opens pre-drafted with the counter-redlined contract attached; the lawyer edits it freely and sends with one click. Sending is always a human act.

One queue, several lawyers: coordination built in

A small team working one shared queue fails in predictable ways — two people silently reviewing the same contract, work stuck in someone's inbox while they're on leave, sales interrupting reviews to ask for status. The workflow closes each of these off:

- **One owner per contract, always.** New arrivals are auto-assigned round-robin, weighted by each reviewer's live load and the contract's tier; a reviewer can also claim from the queue. Opening a colleague's contract is read-only by default, with an explicit take-over — they are notified, and the full audit trail travels — so nobody duplicates work without knowing it.
- **Handoffs keep the context.** Reassignment (holiday cover, load balancing, specialism) moves the whole package — the AI analysis, decisions taken so far, notes — not just the document. An out-of-office toggle reassigns a lawyer's open queue automatically.
- **Escalation lanes instead of shoulder-taps.** Clause escalations and playbook proposals route to the playbook owner's queue; the dashboard shows each person's live load, so rebalancing is a glance, not a meeting.
- **Requesters see status without asking.** The salesperson who sent the contract sees its stage (received / in review / with counterparty / done) — which removes the "any update?" interruptions that fragment review time in the first place.

Accuracy engineering: contracts punish wrong more than slow

- **Deterministic before generative.** Tier 1 verification does not ask a model "does this match?" — the template comparison is an exact text diff computed mechanically; the LLM only classifies the *materiality* of diffs the diff engine found.

Machines are perfect at string comparison; judgement is spent only where judgement is needed.

- **A second opinion where it counts.** For the judgement calls (material vs immaterial, accept vs counter), a second, independent pass — a different model or an independently-prompted skeptic — re-scores the decision. Agreement proceeds; **any disagreement downgrades the contract one tier** and flags the clause. Ensemble the decisions, not the whole pipeline: cost stays sane, and model disagreement becomes a routing signal instead of a hidden error.
- Both sit under the existing belt-and-braces: conservative tiering defaults, human sign-off on everything outbound, and permanent random-sample QA (§5, §7).

5. Delivery: a two-week embed, then autonomy earned by data

The first working version is buildable inside a two-week embed with the legal team. But lawyers are professionally sceptical — correctly so — and autonomy is granted per tier on measured agreement, not on enthusiasm. So the plan separates what ships in the fortnight from what unlocks afterwards.

Week 1 — sit with the lawyers; codify the playbook; measure the baseline. -

Shadow live contract reviews and log where the minutes actually go. This is the baseline every later claim gets measured against. - Mine 50–100 historical NDAs and redline threads to draft the playbook; the lawyers correct it and own it. - Agree the tiering rules and the "never auto-clear" list with the team — their thresholds, not mine.

Week 2 — ship assist mode on NDAs; evaluate; hand over. - Stand up the pipeline on live NDAs: lawyers work from AI-prepared packages (verification reports, verdict cards, drafted responses) with full sign-off on everything that leaves the building. - Evaluate against the week-1 baseline: minutes per contract, tiering agreement, suggestion acceptance — did this actually help, by how much, against what. - Hand over so it keeps running after the embed ends: the playbook and thresholds live with Legal and are editable without a developer; the audit trail is on by default (every AI suggestion stores the playbook version, prompt version and model that produced it, so any past decision can be reproduced); a ten-minute weekly health check (agreement rate, escalation mix, playbook gaps) belongs to the team.

After handover — unlocked by data, not by calendar. - Order forms join once NDAs prove out, shadow mode first. - Tier 1 auto-clear only after sustained near-perfect

agreement in the accumulated sign-off data — with random sampling of auto-cleared contracts routed back for full human review, permanently, to detect drift and automation complacency. - **How the spot-check works:** a tunable random slice of AI-cleared contracts (say 1 in 10 to start — the owner sets the rate, and it never reaches zero) is silently re-queued as a *blind* full review: the reviewing lawyer cannot tell it was AI-cleared, so there is nothing to rubber-stamp. Findings feed the escaped-issue rate (§6); any miss becomes a playbook or threshold fix, not just a statistic. - **Longer term, the pattern travels.** Codify positions → triage by risk → human sign-off → blind sampling is not NDA-specific: order forms first, then DPAs and vendor paperwork, then integration into the CLM/e-signature flow so intake and dispatch are fully systematised. Every expansion re-runs the same trust gates; nothing is grandfathered in.

Starting with mutual NDAs is deliberate: highest volume, most standardised, lowest stakes — the fastest path to trust and measurable relief.

6. Success metrics

Every efficiency number is read against the week-1 baseline — "saved time" only counts against a measured before, not a guessed one.

Efficiency (the point): - Turnaround time, intake → returned/signed (overall and at quarter-end peak) - Lawyer minutes per contract, by tier - % of volume resolved in Tier 1/2 vs Tier 3 - Queue depth at quarter-end vs baseline

Quality (the guardrail — reported alongside, always): - Escaped-issue rate: material issues found in ongoing random sampling of AI-cleared contracts (target: zero material misses) - Tiering accuracy: % of contracts a lawyer would have tiered the same way - Suggestion acceptance rate: % of AI-drafted counters sent with no or minor edits

Adoption (the leading indicator): - % of eligible contracts actually routed through the system - Lawyer-reported confidence, checked in regularly

A specific target I would *not* set in advance: "% auto-cleared." That number should be earned from the agreement data, not promised.

7. Risks and mitigations

RISK	MITIGATION
AI misses a genuinely risky clause	Playbook-grounded checking (not open-ended "review this"); conservative tiering; human sign-off until data supports otherwise; permanent random-sample QA
Confidentiality / privilege	Enterprise LLM deployment with no training on our data; contractual DPA; access controls; counterparty-confidentiality check before any pilot
Lawyers don't adopt it	They own the playbook and the tiering thresholds; system positioned as first-pass associate, not replacement; shadow mode proves value before asking for trust
Automation complacency (rubber-stamping Tier 1)	Sampling injects full-review contracts unannounced; sign-off UI shows the deviation report, not just an "approve" button
Playbook goes stale	Recurring "novel" edits trigger proposed playbook additions; quarterly review owned by legal
Scope creep to complex contracts	Explicit contract-type allowlist; everything else is Tier 3 by default

8. Build vs buy

This market has mature vendors (Spellbook, LegalOn, Luminance, Ironclad and others), so a short vendor scan is honest diligence — worth a day or two, not a quarter. But for a security-conscious, engineering-strong company that wants its contracts to stay inside its own walls and its tooling to fit its own workflow, the thin internal build — LLM API + playbook + a simple review queue, essentially what the demo sketches — is the natural fit: assist mode inside the two-week embed, full control over data handling, and no vendor dependency on the asset that actually matters (the playbook, which transfers either way). Buying remains the honest fallback if, after handover, nobody wants to own even a thin internal tool.

9. What success looks like in one quarter

The legal team stops reading clean NDAs line-by-line. Standard redlines come to them pre-analysed with drafted responses. The quarter-end spike is absorbed by the machine handling volume while humans handle judgement. And the team can show, with

sampling data, that quality did not degrade — which is what makes tier-by-tier expansion defensible to the General Counsel.

Also provided as a plain-text Markdown source (1-solution-design.md) in the same folder.